

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 5: Acids, Bases and Salts

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	5 — Acids, Bases and Salts
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic P4 traps: confusing the colour an indicator shows with the family it names (china rose magenta=acid not base, methyl orange red=acid, turmeric red=base, phenolphthalein pink=base); reading one indicator as conclusive when a second contradicts or merely fails to distinguish (phenolphthalein/turmeric staying unchanged cannot separate acid from neutral); treating 'no litmus change' as proof of water rather than mere neutrality; over- and under-shooting an endpoint during drop-by-drop neutralisation; choosing the wrong member of the acid-base pair to fix a sting, soil, effluent or stomach; assuming the salt from every neutralisation is common salt; and confusing strength or amount of taste with chemical family. Do not write on this paper — mark answers on your answer sheet.

1. A laboratory has four unlabelled colourless liquids known to be lemon juice, sugar water, lime water and soap solution, in some unknown order. A single strip of blue litmus is dipped into each and turns red in exactly one of them. That one liquid must be:

- (A) lemon juice (B) lime water
(C) soap solution (D) sugar water

2. Among curd, milk of magnesia, lime water and vinegar, suppose you remove every substance that turns blue litmus red. The substances that remain are:

- (A) curd and vinegar (B) vinegar and lime water
(C) milk of magnesia and lime water (D) only milk of magnesia

3. A solution turns red litmus blue. A second sample turns blue litmus red. They are mixed in exactly neutralising amounts. Predicting the litmus result of the final mixture, then of adding one extra drop of the second sample, gives:

- (A) first no change in either litmus, then it turns blue litmus red
- (B) first turns red litmus blue, then no change
- (C) first no change, then it turns red litmus blue
- (D) no change in both cases, because mixing cancels everything permanently

4. A student finds a liquid that leaves both red and blue litmus unchanged and also leaves phenolphthalein colourless. The student concludes it is distilled water. The most accurate evaluation is:

- (A) the tests only prove the liquid is neutral; salt water and sugar water would behave identically, so it need not be distilled water
- (B) correct, because only distilled water leaves both litmus and phenolphthalein unchanged
- (C) wrong, because distilled water turns phenolphthalein pink
- (D) wrong, because distilled water turns blue litmus red

5. Dry turmeric paper is touched in turn with soap solution, then lemon juice, then plain water. Recording the colours in that order gives:

- (A) yellow, red, yellow
- (B) red, red, yellow
- (C) red, yellow, yellow
- (D) yellow, yellow, red

6. A drop of china rose indicator is added to vinegar and separately to soap solution. Recalling that china rose is dark pink (magenta) in acids and green in bases, the two colours seen are:

- (A) green in vinegar and magenta in soap solution
- (B) magenta in both, because china rose is always pink
- (C) magenta in vinegar and green in soap solution
- (D) colourless in vinegar and pink in soap solution, like phenolphthalein

7. Phenolphthalein is added to four liquids: tamarind water, lime water, sugar solution and milk of magnesia. The number of these that turn phenolphthalein pink is:

- (A) 1
- (B) 3
- (C) 2
- (D) 0

8. Methyl orange is red in acids and yellow in bases. A drop is added to a fresh sample of lime water, and a separate drop to dilute hydrochloric acid. The colours are, respectively:

- (A) yellow and red
- (B) red and yellow
- (C) pink and colourless
- (D) yellow and yellow

9. A juice tastes very sharply sour, far more sour than a lemon. A student argues this extreme sourness means it has become a base. The correct judgement is:

- (A) correct; very strong sourness flips a liquid into a base
- (B) correct; only mildly sour things are acids

(C) wrong; the strong sourness means it is now neutral

(D) wrong; a stronger sour taste is still an acid property and greater intensity does not change the family

10. Dilute hydrochloric acid is exactly neutralised by sodium hydroxide. Separately, dilute sulphuric acid is exactly neutralised by sodium hydroxide. After each mixture is evaporated, the solids left behind are:

(A) common salt (sodium chloride) in both cases

(B) sodium hydroxide in both cases, since that base was common to both

(C) no solid at all, because neutralisation destroys all matter

(D) two different salts, sodium chloride from the first and sodium sulphate from the second

11. A pink, phenolphthalein-treated soap solution is being titrated by adding lemon juice drop by drop. The pink has just vanished after, say, twenty drops. A student then adds five more drops of lemon juice. The colour now is:

(A) pink again, because more liquid restores the colour

(B) pink again, because excess acid turns phenolphthalein pink

(C) green, because too much acid was added

(D) still colourless, because the extra acid keeps the solution non-basic

12. Three liquids are tested. P turns red litmus blue. Q leaves both litmus papers unchanged. R turns blue litmus red. You then test P, Q and R with turmeric paper. Only one turns turmeric red. It is:

(A) Q

(B) P

(C) R

(D) both P and R

13. A chemist neutralises nitric acid with potassium hydroxide. From the pattern of neutralisation, besides water the product is best described as:

(A) common salt (sodium chloride), as in every neutralisation

(B) more nitric acid, regenerated by the reaction

(C) a salt, formed from the metal part of the base and the acid part of the acid

(D) hydrogen gas and the leftover base

14. A pinch of baking soda is dropped into a glass of vinegar and fizzes; the same pinch dropped into a glass of pure water barely changes anything. The best explanation of the difference is:

(A) vinegar is a base and water is an acid, so only vinegar reacts

(B) baking soda is an acid that fizzes only with the acidic water

(C) baking soda (a base) reacts with the acid in vinegar, releasing a gas, but water is neutral so there is little reaction to release gas

(D) the fizz is just air leaving the baking soda, regardless of the liquid

15. A child is stung by a red ant (which injects formic acid) and, a day later, by a wasp (whose sting is alkaline/basic). The correct single-line first-aid for the two stings is, respectively:

- (A) mild vinegar for the ant sting and baking soda paste for the wasp sting
- (B) baking soda paste for both stings
- (C) baking soda paste for the ant sting and mild vinegar for the wasp sting
- (D) lemon juice for the ant sting and baking soda paste for the wasp sting

16. Among the four indicators litmus, turmeric, china rose and phenolphthalein, suppose you keep only those obtained from a living/plant source. The one you discard is:

- (A) litmus
- (B) phenolphthalein
- (C) turmeric
- (D) china rose

17. A gardener's bed A is too acidic for its crop, while bed B is already too basic. The gardener has only slaked lime. Spreading it on both beds would:

- (A) improve bed A by neutralising its acidity but worsen bed B by making it even more basic
- (B) improve both beds, because lime is neutral and balances any soil
- (C) worsen bed A and improve bed B, because lime is acidic
- (D) have no effect on either, because lime is a salt

18. Two cups, one of dilute acid and one of soap solution, are each touched with turmeric paper; then a few drops of soap solution are slowly stirred into the acid cup until turmeric just begins to turn red there too. At that final moment the acid cup has become:

- (A) still acidic, because acids are stronger than soap
- (B) exactly neutral, since turmeric red marks the neutral point
- (C) a fresh stronger acid, because mixing two liquids strengthens the acid
- (D) basic, since enough base was added to overshoot neutral and now turn turmeric red

19. A liquid is reported to turn phenolphthalein pink yet leave both blue and red litmus papers completely unchanged. The most reasonable conclusion is:

- (A) the liquid is a special neutral substance, since litmus did not change
- (B) the liquid is an acid, because phenolphthalein went pink
- (C) the liquid is salt water, which behaves this way
- (D) the report is inconsistent, because anything basic enough to turn phenolphthalein pink would also turn red litmus blue

20. Solution 1 turns phenolphthalein pink; solution 2 leaves phenolphthalein colourless and turns blue litmus red; solution 3 leaves phenolphthalein colourless and leaves both litmus papers unchanged. The families of 1, 2 and 3 are:

- (A) acid, base, neutral
- (B) base, neutral, acid
- (C) base, acid, neutral
- (D) acid, neutral, base

21. A bottle of stomach antacid syrup is a mild base. Tracing what happens when a spoon of it meets the excess hydrochloric acid in an acidic stomach gives the chain:

- (A) the mild base adds to the acid, making the stomach more acidic and easing the burn
- (B) the syrup coats the stomach with sugar so the acid simply cannot be tasted
- (C) the syrup turns the stomach acid into a stronger acid that the body then expels
- (D) the mild base neutralises the excess acid, producing salt and water, so the burning eases

22. A liquid turns blue litmus red. Predicting what red litmus, phenolphthalein and turmeric will each do with the same liquid gives:

- (A) red litmus stays red, phenolphthalein stays colourless, turmeric stays yellow
- (B) red litmus turns blue, phenolphthalein turns pink, turmeric turns red
- (C) red litmus stays red, phenolphthalein turns pink, turmeric stays yellow
- (D) red litmus turns blue, phenolphthalein stays colourless, turmeric turns red

23. A fizzy soft drink contains dissolved carbon dioxide, which forms carbonic acid. Predicting how it affects blue litmus and china rose gives:

- (A) red litmus turns blue and china rose turns green
- (B) neither litmus changes and china rose keeps its colour
- (C) blue litmus turns red and china rose turns green
- (D) blue litmus turns red and china rose turns dark pink (magenta)

24. Acid rain forms mainly from sulphur dioxide and nitrogen oxides dissolving in rain. A marble statue (calcium carbonate) stands in such rain. The correct chain explaining its slow damage is:

- (A) the rain becomes basic and reacts with the acidic marble, dissolving it
- (B) the dissolved gases make the rain acidic, and that acid reacts with the basic calcium carbonate, eating the marble away
- (C) the rain stays neutral and merely washes loose dust from the surface
- (D) the marble turns acidic in the rain and reacts with itself

25. A chemist mixes equal neutralising amounts of an acid and a base and notices the beaker grows warm. Asked whether evaporating the mixture would then give a cool or warm clue about the salt, the best statement about the warmth's cause is:

- (A) the warmth comes from the acid evaporating, which always heats a beaker
- (B) the warmth comes from the neutralisation reaction itself, which releases heat
- (C) the warmth comes from the salt forming a stronger acid
- (D) the warmth comes from sunlight on the glass, not the reaction

26. A drop of the same china rose indicator is added to cup A (turns magenta) and cup B (turns green). You then add phenolphthalein to fresh portions of A and B. The phenolphthalein colours are:

(A) pink in A and colourless in B

(B) colourless in A and pink in B

(C) pink in both A and B

(D) colourless in both A and B

27. Lemon juice is added drop by drop to a beaker of lime water that contains phenolphthalein and is currently pink. After many drops the pink just disappears; the experimenter keeps adding lemon juice. Testing the final mixture with blue litmus would now show:

(A) no change, because the mixture is locked at neutral

(B) red litmus turns blue, because lime water is always basic

(C) blue litmus turns red, because the extra acid has made the mixture acidic

(D) blue litmus turns green, a sign of overshoot

28. A pinch of crushed chalk (calcium carbonate) is added to two beakers: one of dilute hydrochloric acid, one of soap solution. Predicting where fizzing is seen gives:

(A) fizzing in the soap-solution beaker only, where the base reacts with chalk

(B) fizzing in the acid beaker only, where the carbonate reacts with the acid to release gas

(C) fizzing in both beakers equally

(D) no fizzing in either, because chalk is unreactive

29. A factory's waste water turns blue litmus red. An engineer must make it safe before it reaches a river full of fish. The single correct treatment, and the reason, is:

(A) add more acid, because acid plus acid cancels out

(B) add common salt, because salt soaks up the acidity

(C) add sugar, because dissolving sugar makes any water safe

(D) add a base such as slaked lime, because the waste is acidic and a base neutralises it

30. Methyl orange (red in acids, yellow in bases) is added to a beaker, where it shows red. A student then adds sodium hydroxide solution drop by drop, stirring, until the colour just becomes yellow. The yellow colour signals that:

(A) enough base has been added to neutralise the acid and push the mixture into the basic range

(B) the original beaker held a base that has now turned into an acid

(C) the methyl orange has run out and lost its colour

(D) the mixture is now exactly neutral, the point at which methyl orange is yellow

31. Among the kitchen items curd, lime water, washing soda solution, tamarind water and milk, the number that are bases is:

(A) 2

(B) 1

(C) 3

(D) 4

32. A label-less liquid feels soapy on the fingers, turns turmeric paper red and turns phenolphthalein pink, but the tester refuses to taste it. The safest single classification is:

- (A) an acid, because soapy feel is an acid property
- (B) neutral, because three different tests can never agree
- (C) a base, since soapy feel, turmeric red and phenolphthalein pink all point the same way
- (D) impossible to classify without tasting it

33. Four substances are to be sorted: vinegar, sodium hydroxide, common salt water and milk of magnesia. Grouped as (acids) / (bases) / (neutral), the correct grouping is:

- (A) vinegar and common salt water / sodium hydroxide / milk of magnesia
- (B) sodium hydroxide / vinegar and milk of magnesia / common salt water
- (C) milk of magnesia / vinegar / sodium hydroxide and common salt water
- (D) vinegar / sodium hydroxide and milk of magnesia / common salt water

34. A child rubs honey on a bee sting (which injects an acid) expecting relief, because honey is sweet. The best assessment is:

- (A) honey is a base, so its sweetness neutralises the acid
- (B) any sweet substance neutralises an acid, so honey works
- (C) the sting is basic, so the sweet honey is the right acid for it
- (D) honey will not neutralise the acidic sting; a mild base such as baking soda paste is needed

35. A solution leaves phenolphthalein colourless. A student insists this proves the solution is an acid. The most accurate correction is:

- (A) correct; colourless phenolphthalein always means an acid
- (B) wrong; colourless phenolphthalein always means a base
- (C) wrong; colourless phenolphthalein always means neutral
- (D) colourless phenolphthalein only proves the solution is not a base; it could be an acid or neutral, so a litmus test is needed to decide

36. Pure water is described as neutral. The single cleanest piece of evidence, distinguishing it from both acids and bases at once, is that water:

- (A) turns blue litmus red, like a weak acid
- (B) leaves both red and blue litmus papers unchanged
- (C) turns red litmus blue, like a weak base
- (D) turns phenolphthalein faintly pink

37. A solution turns red litmus blue. It is then mixed bit by bit with dilute hydrochloric acid until red litmus stays red and blue litmus also stays blue. At that point the mixture is:

- (A) still basic, because the original base can never be fully cancelled
- (B) neutral, because just enough acid has neutralised the base so neither litmus changes
- (C) now acidic, because any added acid makes a mixture acidic
- (D) a fresh base stronger than the first

38. A bottle's contents turn blue litmus red, leave phenolphthalein colourless and turn china rose dark pink (magenta). Reading all three results together, the contents are:

- (A) an acid, because all three results agree on acidity
- (B) a base, because china rose changed colour at all
- (C) neutral, because phenolphthalein stayed colourless
- (D) contradictory, because three indicators were used

39. Ammonia solution (used in some glass cleaners) turns red litmus blue. Tracing what the same solution does to turmeric paper and to china rose gives:

- (A) turmeric stays yellow and china rose turns magenta
- (B) turmeric turns red and china rose turns green
- (C) turmeric turns red and china rose turns magenta
- (D) turmeric stays yellow and china rose turns green

40. A monument-conservation team is told that acid rain is corroding a limestone wall (calcium carbonate). They propose periodically washing the wall with a very dilute basic wash. The chemical sense of this plan is:

- (A) unsound, because adding a base to a base (the stone) speeds the damage
- (B) sound, because a base can neutralise the acidic residue left by acid rain on the basic stone
- (C) unsound, because only more acid can clean acid off stone
- (D) pointless, because limestone is acidic and unaffected by any wash

41. Orange juice, soap solution and sugar water are each tested with blue litmus, then with phenolphthalein. The substance that turns blue litmus stay blue AND turns phenolphthalein pink is:

- | | |
|-------------------|------------------|
| (A) orange juice | (B) sugar water |
| (C) soap solution | (D) none of them |

42. A child mixes lemon juice into baking soda solution little by little. Bubbling occurs, then stops, and at that moment neither litmus paper changes. Two more drops of lemon juice are added. The mixture is now:

- (A) still exactly neutral, because the reaction is finished
- (B) slightly acidic, because the extra acid is now in excess after neutralisation
- (C) basic, because baking soda is always the stronger one
- (D) unchanged because lemon juice cannot affect a neutral mixture

43. A teacher claims: 'If you neutralise hydrochloric acid with sodium hydroxide you get common salt, so neutralising any acid with any base also gives common salt.' The correct response is:

- (A) correct; all neutralisations give common salt and water
- (B) wrong; neutralisation never makes common salt

(C) correct; the salt is always common salt but the water differs

(D) wrong; only that particular pair gives sodium chloride, while other acid-base pairs give different salts

44. Why can a single phenolphthalein test sometimes mislead, so that a second indicator is needed?

(A) phenolphthalein is colourless in both acids and neutral solutions, so a colourless result alone cannot separate the two

(B) phenolphthalein turns the same pink in acids and bases, so it cannot tell them apart

(C) phenolphthalein turns red in acids and red in bases, giving no information

(D) phenolphthalein changes colour only in neutral solutions, never in acids or bases

45. A natural indicator and a synthetic indicator are to be paired so each entry is one of each kind. The correct pairing is:

(A) turmeric (natural) and phenolphthalein (synthetic)

(B) litmus (natural) and china rose (synthetic)

(C) phenolphthalein (natural) and turmeric (synthetic)

(D) litmus (natural) and turmeric (synthetic)

46. A beaker of dilute sulphuric acid is slowly neutralised with sodium hydroxide while warm. A student notices the beaker getting warmer and asks if this proves the acid is 'boiling away'. The correct explanation is:

(A) no; the warmth is heat released by the neutralisation reaction, not the acid boiling away

(B) yes; the acid is boiling away, which always heats the beaker

(C) no; the warmth means the base is freezing the mixture

(D) yes; the salt being formed boils off and heats the glass

47. A pond used for fish has become acidic after rain washed acidic matter into it. A worker plans to spread a base over the water. Which choice is both a base and a cheap, commonly used one for this purpose?

(A) vinegar

(B) lemon juice

(C) slaked lime

(D) more of the acidic runoff

48. Three liquids X, Y, Z are tested. X turns blue litmus red. Y turns red litmus blue. Z changes neither litmus. Which single liquid, treated with methyl orange (red in acids, yellow in bases), turns it red?

(A) X

(B) Y

(C) Z

(D) both X and Y

49. A student lists tests of an unknown liquid: it tastes sour, feels neither soapy nor slippery, turns blue litmus red, leaves phenolphthalein colourless and turns china rose magenta. The single best conclusion is:

- (A) the liquid is a base, because china rose changed colour
- (B) the liquid is neutral, because phenolphthalein stayed colourless
- (C) the liquid is an acid, since every test points to acidity
- (D) the results conflict, so no conclusion is possible

50. A solution turns china rose green. Predicting how the same solution behaves with red litmus, with phenolphthalein and to the touch gives:

- (A) red litmus stays red, phenolphthalein stays colourless, and it tastes sour
- (B) red litmus turns blue, phenolphthalein turns pink, and it feels soapy
- (C) blue litmus turns red, phenolphthalein turns pink, and it feels soapy
- (D) red litmus turns blue, phenolphthalein stays colourless, and it tastes sour

51. A teacher writes 'acid + base $\rightarrow$ salt + water' and asks for the best everyday example of this exact reaction. The strongest example is:

- (A) dissolving sugar in tea
- (B) ice melting in a glass of juice
- (C) a slice of apple turning brown in air
- (D) swallowing an antacid (a base) to relieve excess stomach acid

52. An acidic solution is neutralised by adding exactly the right amount of base, then the water is boiled off. A student claims the dry solid left behind is the leftover acid. The correct statement is:

- (A) the solid is the leftover acid, which dried out
- (B) the solid is the leftover base, which never reacts fully
- (C) the solid is the salt formed in the neutralisation, since no acid remains after exact neutralisation
- (D) there is no solid, because neutralisation removes all matter

53. A list reads: vinegar — acid, lime water — base, milk of magnesia — base, soda water — base, lemon juice — acid. Exactly one pairing is wrong. It is:

- (A) vinegar — acid
- (B) soda water — base
- (C) lime water — base
- (D) milk of magnesia — base

54. A liquid turns blue litmus red but a student wants to be doubly sure it is an acid before reporting it. Which second result would best confirm it is acidic (and not, say, a reporting error)?

- (A) china rose turns dark pink (magenta) in it
- (B) phenolphthalein turns pink in it
- (C) turmeric paper turns red in it
- (D) red litmus turns blue in it

55. Why is it unsafe to conclude 'this liquid is pure water' merely because it changes neither litmus paper?

- (A) many neutral liquids besides water, such as salt water and sugar water, also change neither litmus paper
- (B) because pure water actually turns blue litmus red, so no change means it is not water
- (C) because only acids and bases leave litmus unchanged, never water
- (D) because litmus cannot be used on any liquid that is colourless

56. A neutralisation is carried out in which lemon juice is added to lime water. Asked which product makes the reaction 'acid + base $\rightarrow$ salt + water', a student must identify the salt and the water. The correct identification of where the water comes from is:

- (A) the water is just the water the lime water was dissolved in, and no new water forms
- (B) the water is the leftover lemon juice
- (C) there is no water in the products, only salt
- (D) the water is a product newly formed when the acid and base react, alongside the salt

57. A liquid turns blue litmus red. Asked whether it could be soap solution, distilled water, vinegar or milk of magnesia, the only one that fits is:

- (A) soap solution
- (B) distilled water
- (C) vinegar
- (D) milk of magnesia

58. A student tests four liquids with red litmus: ammonia solution turns it blue, vinegar leaves it red, lime water turns it blue, distilled water leaves it red. The number of bases identified is:

- (A) 1
- (B) 3
- (C) 4
- (D) 2

59. Two unknowns are tested with the same china rose indicator: cup P turns magenta, cup Q turns green. You then need to decide which cup, when slowly added to the other, would behave as the acid in a neutralisation. The acid is:

- (A) cup Q, because china rose green marks an acid
- (B) cup P, because china rose magenta marks an acid
- (C) cup P, because magenta always means a base
- (D) neither, because china rose cannot tell acids from bases

60. A student neutralises an acid with a base and, just as both litmus papers stop changing, declares the job done. To check that the endpoint was reached without overshooting, the single most reliable confirmation is that:

- (A) the mixture turns blue litmus red, showing the base has been fully used up
- (B) blue litmus stays blue and red litmus stays red, showing the mixture is neutral with no excess of either
- (C) the mixture turns red litmus blue, showing the acid has been fully used up
- (D) the mixture turns turmeric red, the colour seen at a perfect neutral point

Solutions — Science (Class 7), Chapter 5: Acids, Bases and Salts — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	D	21	D	31	A	41	C	51	D
2	C	12	B	22	A	32	C	42	B	52	C
3	A	13	C	23	D	33	D	43	D	53	B
4	A	14	C	24	B	34	D	44	A	54	A
5	C	15	C	25	B	35	D	45	A	55	A
6	C	16	B	26	B	36	B	46	A	56	D
7	C	17	A	27	C	37	B	47	C	57	C
8	A	18	D	28	B	38	A	48	A	58	D
9	D	19	D	29	D	39	B	49	C	59	B
10	D	20	C	30	A	40	B	50	B	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Blue litmus turns red only in an acid. Of the four, only lemon juice (citric acid) is an acid, so it is the single liquid that turns blue litmus red. Lime water and soap solution are both bases, which leave blue litmus unchanged (they turn red litmus blue instead). Sugar water is neutral and leaves blue litmus unchanged. Since exactly one strip turned red, that liquid is the acid, i.e. lemon juice.

2. (C) Turning blue litmus red is an acid property, so removing every such substance removes all the acids. Curd (lactic acid) and vinegar (acetic acid) are acids, so they are removed. Milk of magnesia (magnesium hydroxide) and lime water are bases, which do not turn blue litmus red, so both remain. Keeping only milk of magnesia would wrongly discard the equally basic lime water; keeping vinegar would keep an acid.

3. (A) The first solution is a base (red litmus to blue) and the second is an acid (blue litmus to red). Mixed in exactly neutralising amounts they cancel, so the mixture is neutral and changes neither litmus paper. Adding one extra drop of the second sample (the acid)

tips the mixture slightly acidic, so it now turns blue litmus red. It would not turn red litmus blue (that needs excess base), and neutrality is not permanent once more acid is added.

4. (A) Leaving both litmus papers unchanged and phenolphthalein colourless are exactly the results of any neutral liquid. Distilled water is neutral, but so are common salt water and sugar water, which would give identical results. So the tests establish neutrality, not identity; the liquid need not be distilled water. Distilled water does not turn phenolphthalein pink (only a base does) and does not turn blue litmus red (only an acid does).

5. (C) Turmeric turns red only in a base and stays yellow in acids and in neutral liquids. Soap solution is basic, so the paper turns red. Lemon juice is acidic, so the paper stays yellow. Plain water is neutral, so the paper stays yellow. The order is therefore red, yellow, yellow. Reversing red to the acid (lemon) misremembers turmeric's direction; turmeric does not turn red in acids or in water.

6. (C) China rose shows dark pink (magenta) in acids and green in bases. Vinegar is acidic (acetic acid), so it turns china rose magenta. Soap solution is basic, so it turns china rose green. Swapping the two reverses china rose's direction. China rose is not a single fixed colour, and it is not colourless like phenolphthalein in acid; it goes magenta there.

7. (C) Phenolphthalein turns pink only in a base. Lime water and milk of magnesia are bases, so both turn it pink, giving a count of 2. Tamarind water is acidic (tartaric acid) and sugar solution is neutral, so both leave phenolphthalein colourless. Counting 1 forgets that both lime water and milk of magnesia are bases; counting 3 wrongly includes an acid or neutral.

8. (A) Methyl orange is yellow in bases and red in acids. Lime water is a base, so methyl orange is yellow in it; dilute hydrochloric acid is an acid, so methyl orange is red in it. The order is yellow then red. Reversing them confuses the directions; methyl orange shows neither pink nor colourless (those are phenolphthalein behaviours), and the acid would not also be yellow.

9. (D) Sour taste is the signature of an acid no matter how weak or strong it is; a sharper sour taste simply means a more strongly acidic liquid, not a base. Intensity of a taste does not switch the chemical family. A base would taste bitter, not sour, and a neutral liquid has no sharp taste, so neither of those follows from extreme sourness.

10. (D) Neutralisation gives a salt plus water, and the salt depends on which acid and base reacted. Hydrochloric acid with sodium hydroxide gives sodium chloride; sulphuric acid with sodium hydroxide gives sodium sulphate. So the two solids are different. They are not both common salt, since the acids differ. None of the base remains after exact neutralisation, and matter is not destroyed, so a solid salt does form in each.

11. (D) Phenolphthalein is pink only in a base and colourless otherwise. When the pink vanished, the base had been neutralised and the solution was about neutral. Adding more lemon juice (acid) makes the solution acidic, which is also colourless for phenolphthalein, so it stays colourless. Adding more liquid does not bring back pink, excess acid never makes phenolphthalein pink, and green is not a phenolphthalein colour.

12. (B) Turmeric turns red only in a base. P turns red litmus blue, so P is a base and turns turmeric red. Q changes neither litmus, so Q is neutral and leaves turmeric yellow. R turns blue litmus red, so R is an acid and leaves turmeric yellow. Hence only P turns turmeric red. R, being acidic, does not; Q, being neutral, does not.

13. (C) Every acid-base neutralisation gives a salt and water, and the salt is built from the base's metal part combined with the acid's acidic part. Here that salt is not sodium chloride, because neither sodium nor chlorine is involved; sodium chloride only forms from hydrochloric acid and a sodium base. Neutralisation does not regenerate the starting acid, and hydrogen is released when acids meet metals, not in plain acid-base neutralisation, where no base is left over either.

14. (C) Baking soda is a base; vinegar is an acid, so they undergo a neutralisation that releases carbon dioxide gas, seen as fizzing. Pure water is neutral, so there is no acid for the baking soda to react with and little gas is produced. Vinegar is the acid (not a base) and water is neutral (not an acid). The bubbles are a gas produced by a real reaction, not trapped air, which is why they appear only with the acidic vinegar.

15. (C) Each sting is relieved by neutralising it with the opposite family. The ant sting is acidic, so a mild base, baking soda paste, neutralises it. The wasp sting is basic, so a mild acid, vinegar, neutralises it. Reversing the two would add the same family and worsen each. Baking soda would help the acid sting but worsen the basic wasp sting, and lemon juice is an acid, useless on an acidic ant sting.

16. (B) Phenolphthalein is a synthetic, lab-made indicator, so it is the one discarded when keeping only natural ones. Litmus comes from lichens, turmeric from the turmeric plant, and china rose from hibiscus flowers, so all three are natural and are kept. Discarding any of the three natural indicators would be wrong.

17. (A) Slaked lime is a base. Added to acidic bed A it neutralises the excess acid and improves the soil. Added to already-basic bed B it adds more base, making it even more basic and worsening it. Lime is not neutral, so it does not balance any soil; it is not acidic, so it does not fix the basic bed; and it is not an inert salt, since it clearly changes acidic soil.

18. (D) Turmeric turns red only in a base, so when the acid cup finally turns turmeric red, enough basic soap solution has been added to neutralise all the acid and then push past neutral into basic. It is therefore basic, not still acidic. Turmeric red does not mark the

neutral point (neutral leaves turmeric yellow), and adding a base to an acid neutralises it rather than creating a stronger acid.

19. (D) Phenolphthalein turns pink only in a base, and a base also turns red litmus blue. A liquid cannot be basic for phenolphthalein yet neutral for litmus at the same time, so the two results contradict each other and the report is inconsistent. A neutral substance leaves phenolphthalein colourless, an acid leaves phenolphthalein colourless too (pink means base, not acid), and salt water is neutral, so none of those explain a genuine pink-with-no-litmus-change result.

20. (C) Solution 1 is pink with phenolphthalein, which happens only in a base. Solution 2 is colourless with phenolphthalein and turns blue litmus red, the mark of an acid. Solution 3 is colourless with phenolphthalein and changes neither litmus, so it is neutral. Hence base, acid, neutral. The other orders mislabel the phenolphthalein-pink solution or treat the blue-litmus-to-red solution as something other than an acid.

21. (D) Acidity is too much hydrochloric acid in the stomach. An antacid is a mild base, and a base neutralises an acid to give salt and water, removing the excess acid and easing the burning. A base does not make a stomach more acidic; that would worsen the problem. The relief is chemical neutralisation, not a sugar coating, and an antacid does not strengthen the acid.

22. (A) Blue litmus turning red marks an acid. In an acid red litmus shows no change (it is already red and only a base turns it blue), phenolphthalein stays colourless (it is pink only in a base), and turmeric stays yellow (it turns red only in a base). So all three remain unchanged. The other options apply base behaviours (red litmus blue, phenolphthalein pink, turmeric red) to what is actually an acid.

23. (D) Dissolved carbon dioxide makes carbonic acid, so the drink is mildly acidic. An acid turns blue litmus red, and china rose shows dark pink (magenta) in acids, so both of those happen. The drink is not basic, so red litmus would not turn blue and china rose would not turn green. It is not neutral, so litmus does change. Pairing acid litmus with a green (basic) china rose mixes the directions.

24. (B) Sulphur dioxide and nitrogen oxides dissolve to form acids, so acid rain is acidic. Marble is calcium carbonate, which behaves as a base, so the acid reacts with and gradually corrodes it. The rain is not basic and the marble is not acidic, so the reversed chain is wrong. The rain is acidic, not neutral, and does more than wash dust; the marble does not react 'with itself'.

25. (B) Acid-base neutralisation releases heat, so the beaker warms up as the reaction proceeds. Evaporation, if anything, takes in heat and cools, so it is not the source of the warmth. Neutralisation produces a salt and water, not a stronger acid, so no acid-strengthening warms the beaker. The heat is from the reaction, not from external sunlight.

26. (B) China rose is magenta in acids and green in bases, so cup A is an acid and cup B is a base. Phenolphthalein is colourless in acids and pink in bases, so A (acid) stays colourless and B (base) turns pink. Reversing this would treat magenta as basic, which is wrong. Both pink would need both to be bases; both colourless would need neither to be basic, but B is.

27. (C) While pink, the lime water was basic. The pink vanishing marks the point where the lemon juice (acid) has neutralised the base, leaving the mixture about neutral. Continuing to add lemon juice makes the mixture acidic, so blue litmus now turns red. The mixture is not locked at neutral once more acid is added; lime water has been used up so it is no longer basic; and litmus has no green result.

28. (B) Chalk is calcium carbonate, which reacts with an acid to release carbon dioxide gas, seen as fizzing. The acid beaker therefore fizzes. Soap solution is basic, and a carbonate does not fizz with a base in this way, so the soap beaker does not fizz. Fizzing in both, or none at all, ignores that the reaction needs an acid. Chalk is not unreactive towards acids.

29. (D) Blue litmus turning red shows the waste is acidic, so it must be neutralised by a base; slaked lime is the standard cheap base used. Adding more acid makes it more acidic, not safer, and two acids do not cancel. Common salt is neutral and does not neutralise an acid. Sugar merely dissolves and has no neutralising action.

30. (A) Methyl orange is red in acids, so the beaker first held an acid. Adding the base sodium hydroxide neutralises the acid, and once enough base is added the mixture becomes basic, where methyl orange is yellow. So yellow signals the mixture has reached the basic range. The original was an acid (red), not a base; the indicator has not run out; and yellow marks basic conditions, not the exact neutral point.

31. (A) Bases turn red litmus blue and feel soapy. Lime water and washing soda solution are bases, giving a count of 2. Curd (lactic acid) and tamarind water (tartaric acid) are acids, and milk is roughly neutral, so none of those three is a base. A count of 1 misses one base, while 3 or 4 wrongly counts an acid or the neutral milk as basic.

32. (C) Soapy feel, turmeric turning red and phenolphthalein turning pink are all base properties, and they agree, so the liquid is a base. Soapy feel is a base trait, not an acid one. A neutral substance would change none of these indicators, so three positive base results rule out neutral. Tasting is unnecessary and unsafe; the indicator results already give a confident answer.

33. (D) Vinegar (acetic acid) is the acid. Sodium hydroxide and milk of magnesia (magnesium hydroxide) are both bases. Common salt water is neutral. So acids = vinegar, bases = sodium hydroxide and milk of magnesia, neutral = common salt water. Salt water

is not an acid, sodium hydroxide is not the lone base, and milk of magnesia is not neutral, so the other groupings misplace at least one substance.

34. (D) A bee sting is acidic and is relieved by neutralising it with a mild base such as baking soda paste. Honey is sweet but not a base, so it does not neutralise the acid. Sweetness has nothing to do with neutralising power, so sweet substances in general do not neutralise acids. The bee sting is acidic, not basic, so it does not need an acid, and honey is not an acid remedy anyway.

35. (D) Phenolphthalein is pink in bases and colourless in both acids and neutral solutions. So a colourless result only rules out a base; it cannot by itself tell an acid from a neutral substance. A second test, such as blue litmus, is needed: if it turns red the solution is an acid, otherwise neutral. Colourless does not 'always' mean acid, base or neutral, since it covers two of the three families at once.

36. (B) A neutral substance changes neither litmus paper, and water does exactly that, distinguishing it from acids (which turn blue litmus red) and bases (which turn red litmus blue) in one test. Turning blue litmus red would make water an acid; turning red litmus blue would make it a base; turning phenolphthalein pink would also mark a base. Only 'no change in either litmus' fits a neutral liquid.

37. (B) The original solution turned red litmus blue, so it was a base. Adding the acid neutralises it, and when neither litmus changes the mixture has reached neutral, where the acid and base have cancelled. It is not still basic, since red litmus no longer stays blue would otherwise show. It is not yet acidic, because that needs excess acid (blue litmus would then turn red). Neutralisation does not create a stronger base.

38. (A) Blue litmus turning red is an acid sign, phenolphthalein staying colourless is consistent with an acid (it is pink only in a base), and china rose magenta is the acid colour. All three agree on acidity, so the contents are an acid. China rose changing to magenta is itself the acid result, not a base sign. Colourless phenolphthalein alone could be neutral, but the litmus change rules that out, so the results are consistent, not contradictory.

39. (B) Red litmus turning blue shows ammonia solution is a base. A base turns turmeric red and turns china rose green. So turmeric is red and china rose is green. Turmeric staying yellow is the acid/neutral result, not a base one. China rose magenta is the acid colour, the opposite of what a base gives. Only 'turmeric red and china rose green' matches a base on both counts.

40. (B) Acid rain leaves acidic residue, and a base neutralises an acid, so a dilute basic wash can neutralise that residue and slow further attack on the calcium-carbonate stone. The stone being a base does not mean a base wash 'attacks' it; the wash targets the acid

residue, not the stone. Adding more acid would worsen corrosion, and limestone is a base (calcium carbonate), not acidic, so it is certainly affected by acid rain.

41. (C) Leaving blue litmus blue (no change) plus turning phenolphthalein pink are both base behaviours; only a base does both. Soap solution is the base, so it fits. Orange juice is acidic (citric acid): it turns blue litmus red and leaves phenolphthalein colourless. Sugar water is neutral: blue litmus stays blue but phenolphthalein stays colourless, so it fails the phenolphthalein test. Hence soap solution is the only match.

42. (B) When bubbling stops and neither litmus changes, the acidic lemon juice and basic baking soda have just neutralised, leaving a neutral mixture. Adding two more drops of lemon juice puts acid in excess, making the mixture slightly acidic. It is not still exactly neutral once extra acid is added; baking soda has been used up so it is not basic; and lemon juice (an acid) certainly does affect a neutral mixture by making it acidic.

43. (D) Neutralisation always gives a salt and water, but the salt depends on the specific acid and base. Hydrochloric acid with sodium hydroxide gives sodium chloride (common salt), but, say, sulphuric acid with sodium hydroxide gives sodium sulphate instead. So the general claim is wrong; different pairs give different salts. Neutralisation can make common salt (from the right pair), and it is the salt that varies, not the water.

44. (A) Phenolphthalein is pink only in a base and colourless in both acids and neutral solutions. So a colourless reading could mean either an acid or a neutral substance, and a second indicator such as litmus is needed to decide. Phenolphthalein does not turn pink in acids (only bases), it does not turn red at all, and it does change colour for bases, so the other explanations misstate its behaviour.

45. (A) Phenolphthalein is the synthetic, lab-made indicator; turmeric, litmus and china rose are all natural. So turmeric (natural) with phenolphthalein (synthetic) is one of each. China rose is natural, not synthetic, so litmus-and-china-rose is two naturals. Calling phenolphthalein natural or turmeric synthetic reverses their true kinds, and litmus-and-turmeric are both natural.

46. (A) Neutralisation of an acid by a base releases heat, so the beaker warms up as the reaction proceeds; nothing is boiling away. Acid boiling away would, if anything, need heat put in and would not be caused by the reaction. A base does not freeze the mixture, and the salt being formed stays dissolved rather than boiling off. The warmth is simply the reaction's released heat.

47. (C) The pond is acidic and must be neutralised by a base; slaked lime is a cheap, widely used base for treating acidic water. Vinegar (acetic acid) and lemon juice (citric acid) are acids, so they would worsen the acidity rather than fix it. Adding more acidic runoff obviously makes the problem worse. Only slaked lime fits both 'a base' and 'cheap and common'.

48. (A) Methyl orange is red only in an acid. X turns blue litmus red, so X is an acid and turns methyl orange red. Y turns red litmus blue, so Y is a base, and methyl orange is yellow (not red) in it. Z changes neither litmus, so Z is neutral, and methyl orange stays yellow there too. Hence only X turns methyl orange red; Y is a base where it is yellow.

49. (C) Sour taste, blue litmus turning red, colourless phenolphthalein and china rose magenta are all acid behaviours, and they agree, so the liquid is an acid. China rose changing to magenta is precisely the acid colour, not a base sign. Colourless phenolphthalein alone could be neutral, but the sour taste and blue-litmus-to-red rule that out. The five results are consistent, not in conflict.

50. (B) China rose green is the base colour, so the solution is a base. A base turns red litmus blue, turns phenolphthalein pink and feels soapy, so that combination follows. Red litmus staying red with a sour taste describes an acid, the opposite. Blue litmus turning red is an acid behaviour, which a base does not show. Phenolphthalein staying colourless contradicts a base, which turns it pink.

51. (D) An antacid is a base, and it neutralises stomach acid to give salt and water, exactly matching acid plus base gives salt plus water. Dissolving sugar is mere mixing, not an acid-base reaction. Ice melting is a physical change of state, not a chemical neutralisation. An apple browning is a reaction with air, not between an acid and a base. Only the antacid example is a genuine neutralisation.

52. (C) Exact neutralisation uses up all the acid and base, leaving salt dissolved in the water formed. Boiling off the water leaves that solid salt behind. No acid remains, so it cannot be 'leftover acid'; no base remains either, since the amounts were exact. Matter is not destroyed, so a solid does form. The correct answer is the salt.

53. (B) Soda water contains dissolved carbon dioxide, which forms carbonic acid, so it is an acid, not a base; that is the wrong pairing. Vinegar (acetic acid) is correctly an acid, lime water is correctly a base, and milk of magnesia (magnesium hydroxide) is correctly a base. So the single error is calling soda water a base when it is actually acidic.

54. (A) China rose shows dark pink (magenta) in acids, so a magenta china rose result agrees with blue litmus turning red and confirms the liquid is an acid. Phenolphthalein turning pink would indicate a base, contradicting the acid finding. Turmeric turning red also indicates a base, again contradicting it. Red litmus turning blue is a base behaviour too. Only china rose magenta is a second, independent acid sign.

55. (A) Changing neither litmus paper proves a liquid is neutral, but water is only one of many neutral liquids; salt water and sugar water give the same result. So the test cannot single out pure water. Water does not turn blue litmus red (it is neutral). Acids and bases do change litmus, so they are not the ones leaving it unchanged. Litmus works fine on colourless liquids, so that objection is false.

56. (D) In acid + base $\rightarrow$ salt + water, the water is genuinely a product of the reaction, formed when the acid and base combine, along with the salt. It is not merely the solvent water that was already present; the equation states water as a product. The leftover lemon juice would be excess acid, not water. And water is always one of the two products, so it is wrong to say only salt forms.

57. (C) Blue litmus turning red is the test for an acid. Vinegar (acetic acid) is an acid, so it fits. Soap solution and milk of magnesia are bases, which turn red litmus blue, not blue litmus red. Distilled water is neutral and changes neither litmus paper. So vinegar is the only acidic liquid among the four that turns blue litmus red.

58. (D) Red litmus turning blue is the test for a base. Ammonia solution and lime water turn red litmus blue, so two are bases. Vinegar is an acid and leaves red litmus red; distilled water is neutral and also leaves red litmus red. So exactly 2 of the four are bases. A count of 1 misses one base, while 3 or 4 wrongly counts the acid or neutral liquid as basic.

59. (B) China rose is magenta in acids and green in bases. Cup P (magenta) is therefore the acid, and cup Q (green) is the base; in a neutralisation cup P plays the acid. Saying green marks an acid reverses china rose's direction. Magenta marks an acid, not a base. China rose certainly distinguishes acids from bases, since it shows two different colours, so it is not useless here.

60. (B) At a clean neutral endpoint, neither litmus paper changes: blue stays blue and red stays red, showing no excess acid or base remains. If blue litmus turned red, an excess of acid would be present, meaning the endpoint was overshoot with too much acid. If red litmus turned blue, an excess of base would remain, an overshoot the other way. Turmeric turns red only in a base, so turmeric red would actually signal an excess of base, not a perfect neutral point.